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Optimization of Semiconductor Supply Chain Networks

23MNG336 - Operations Research

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Introduction

- A global microchip producer needs to fulfill thousands of customer orders through its outbound logistics network.
- Orders can be routed through multiple warehouses, origin ports, and transportation carriers.
- Each warehouse has limited capacity, product availability, and customer-specific restrictions.
- Transportation options differ in cost, weight-based pricing, service level, and transportation mode.
- The key challenge is to determine cost-effective and feasible logistics decisions while satisfying all customer demand and operational constraints.

Dataset Overview

Supply Chain Logistics Dataset

- Publicly available real-world dataset of a global microchip producer's outbound logistics.
- Covers one month of customer orders across the company's full warehouse and port network.
- Seven related tables describe demand, freight rates, warehouse economics and connectivity.



Figure 1: Structure of the outbound logistics network

- Freight is moved by 9 carriers over air and ground modes, under three service levels (DTD, DTP and CRF), with weight-banded pricing and minimum charges.

Dataset Structure

Dataset Table	Rows	Columns	What It Describes
OrderList	9,215	14	Customer demand: weight, service level, dates
FreightRates	1,540	11	Carrier lane rates by weight band and service
WhCosts	19	2	Storage cost per unit at each warehouse
WhCapacities	19	2	Daily order-handling capacity of each warehouse
ProductsPerPlant	2,036	2	Products each warehouse is able to ship
VmiCustomers	14	2	Customers restricted to specific warehouses
PlantPorts	22	2	Warehouse to origin-port connectivity

Table 1: Size and role of each table in the supply chain logistics dataset

- OrderList alone cannot be optimised: cost, feasibility and capacity information live in the other six tables.

Dataset Relationships

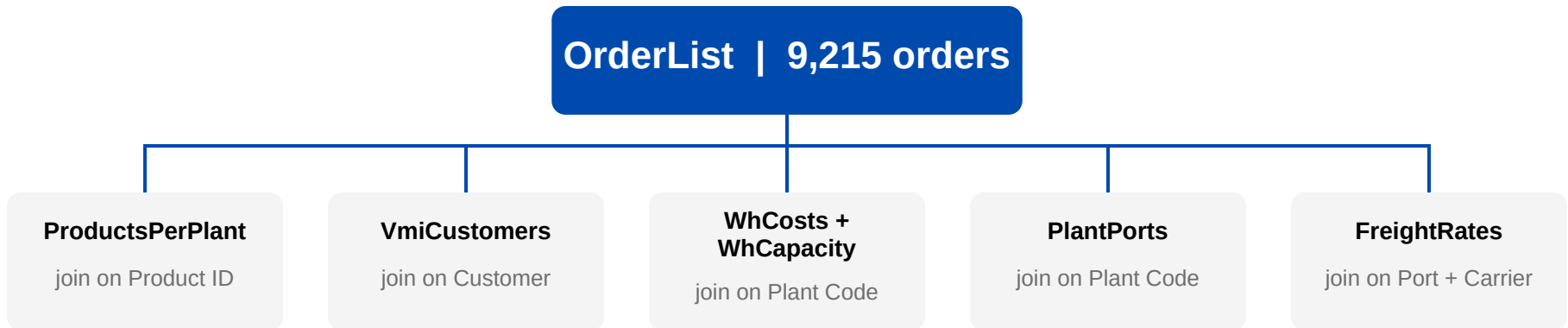


Figure 2: Relationships between the seven dataset tables

- Every join acts as a feasibility filter, removing routes an order is not allowed to take.
- A warehouse is usable only if it stocks the product and is permitted for that customer.
- Only warehouse-port pairs listed in PlantPorts are physically connected.
- Only carrier-lane-service combinations present in FreightRates can be priced and selected.

Scale of the Network

Network Attribute	Value	Source Table
Customer orders to be routed	9,215	OrderList
Distinct customers / products	46 / 772	OrderList
Warehouses (plants)	19	WhCapacities
Origin ports / destination ports	11 / 1	PlantPorts
Warehouse-to-port connections	22	PlantPorts
Carriers available	9	FreightRates
Freight rate lines (weight bands)	1,540	FreightRates
Service levels / transport modes	3 / 2	OrderList

Table 2: Scale of the outbound logistics network

- With 19 warehouses, 11 ports and 9 carriers, an order can have hundreds of feasible routes, so manual selection is not practical.

Problem Statement

Given a set of customer orders and a constrained logistics network, how can we determine the most effective warehouse, origin port, carrier and transportation decisions, so that total logistics cost is minimised while every operational constraint is satisfied?

- Every order must be assigned exactly one warehouse, one origin port and one carrier.
- The assignment must respect capacity, product availability, customer and connectivity restrictions.
- The historical dataset records what was done, not what was optimal.
- The project converts this business problem into an Operations Research optimisation model.

Constraints in the Problem

- Every order must be assigned to exactly one warehouse, origin port and carrier.
- Warehouse daily capacity is limited and varies widely (WhCapacities).
- Not every warehouse stocks every product (ProductsPerPlant).
- Some customers may only be served by specific warehouses (VmiCustomers).
- Not every warehouse is connected to every origin port (PlantPorts).
- Not every carrier operates on every shipping lane (FreightRates).
- Freight cost depends on the weight band into which the shipment falls.
- A minimum freight charge applies when the computed cost falls below it.
- Service levels (DTD, DTP, CRF) and modes (air, ground) differ in cost and transit time.

These restrictions eliminate most warehouse-port-carrier combinations, so the cheapest feasible route cannot be read off the data and must be found by optimisation.

Optimisation Problem

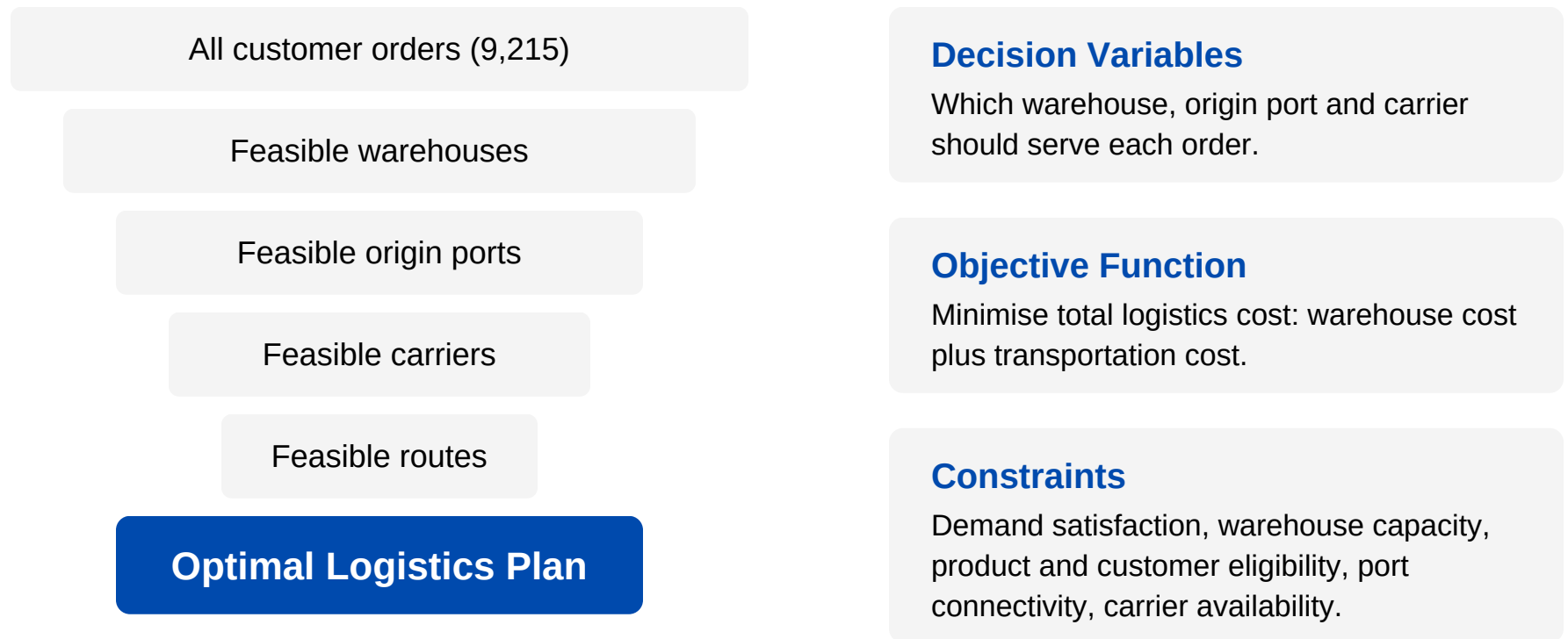


Figure 3: Narrowing the solution space before optimisation

Objective Function

Minimising Overall Logistics Cost

$$\text{Total Cost} = \text{Warehouse Cost} + \text{Transportation Cost}$$

Warehouse Cost

unit quantity $\times$ *cost per unit of the assigned warehouse*

Cost per unit varies from 0.37 to 2.04 across the 19 warehouses, so warehouse choice alone changes the bill.

Transportation Cost

shipment weight $\times$ *rate of the applicable weight band*

Depends on carrier, lane, service level and mode, and is raised to the minimum charge when it falls below it.

Minimum-Cost Logistics Assignment

Which warehouse, origin port and carrier should be assigned to each order so that total logistics cost is minimised?

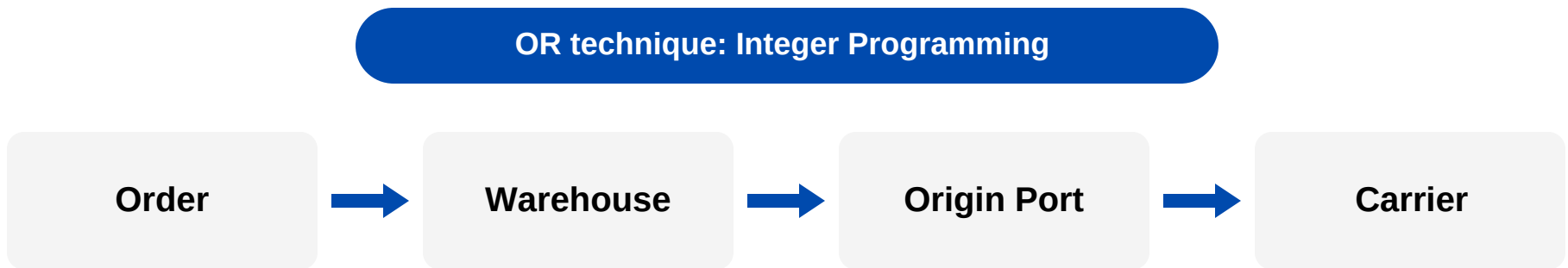


Figure 4: Assignment decision taken for every order

- One binary variable is created for every feasible order-warehouse-port-carrier combination.
- Exactly one combination is selected for each order.
- Each candidate is priced using the correct weight band and minimum freight charge.
- The result is an order-level routing plan that is feasible and least-cost.

Cost versus Delivery Time

How should the company balance transportation cost against delivery time when the two objectives point to different routes?

OR technique: Goal Programming / Multi-Criteria Decision Making

Route A

Lower freight rate, longer transit. Ground and slower service levels can take up to 14 days.

Route B

Higher freight rate, shorter transit. Air and premium service levels deliver in 0 to 3 days.

- Transit times in the freight data range from 0 to 14 days, so the choice is real and measurable.
- A single cost objective would always pick the slow route, which is not acceptable to customers.
- Goal programming lets the user set a target cost and a target delivery time and minimise deviation.

Sensitivity Analysis

If a key parameter changes, does the optimal plan stay optimal, and at what point does the routing decision actually flip?

Warehouse Capacity

+/- 10% and 20%

Freight Rates

+10% and +20%

Warehouse Cost

per-unit cost shifts

Demand Volume

seasonal surges

- What happens to total cost if warehouse capacity is expanded?
- What happens if a carrier raises its rates by 10 or 20 percent?
- At what point does the optimal warehouse, port or carrier choice change?
- Which parameter has the largest influence on the optimal solution?

Technology and Implementation

Layer	Proposed Technology	Purpose
Frontend / UI	React	Upload, controls, dashboards
Data processing	Python with pandas	Cleaning and joining the tables
Optimisation modelling	PuLP	Building IP, LP and GP models
Solver	CBC or HiGHS	Solving the models
Visualisation	Matplotlib	Charts and network views
Backend and storage	FastAPI with SQLite	Serving and storing scenarios

Table 9: Indicative technology stack for the decision support system

Expected Outcomes

- A minimum-cost, fully feasible logistics plan for all 9,215 orders.
- An optimised transportation network showing how demand should flow to the destination port.
- A cost versus delivery-time trade-off analysis under user-selected priorities.
- A sensitivity study showing how robust the optimal plan is to parameter changes.
- An interactive decision-support interface with statistics.
- Visual analytics that logistics managers can read without knowing the mathematics.